

AI Supervisor

Bolt Tightening

Compliance Demo

How BiztelAI detects SOP violations in real time —
without a human supervisor on the line.

4 Bolts

tracked
simultaneously

< 1s

detection
latency

100%

automated
alerts

24/7

continuous
monitoring

What is the BiztelAI AI Supervisor?

BiztelAI's AI Supervisor is a plug-and-play computer vision system that mounts above any assembly station and monitors worker actions in real time. It detects whether bolts are tightened in the correct sequence, flags violations instantly, and logs every cycle — with zero human oversight required.

Computer Vision Tracking

- Camera locks onto each bolt (B1-B4) using edge detection — no markers
- or hardware changes needed

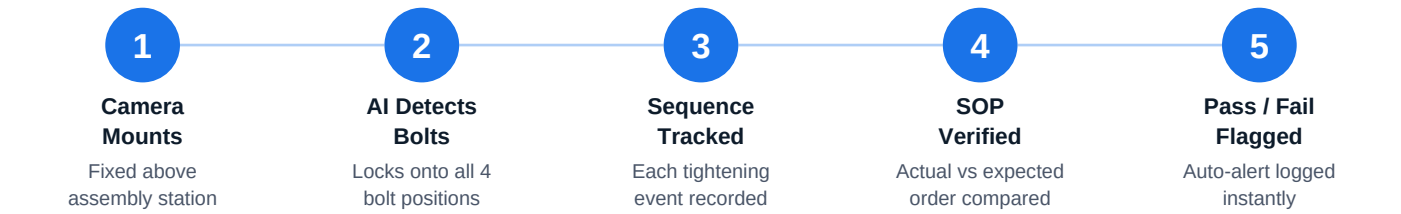
SOP Sequence Verification

- Checks actual bolt order vs SOP-defined expected
- order in real time as the worker tightens each bolt

Instant Violation Alerts

- Red highlight + on-screen alert if wrong bolt selected
- — logged automatically with timestamp and cycle data

HOW IT WORKS — 5-STEP FLOW



DEMO SETUP — WHAT YOU WILL SEE IN THE SCREENSHOTS

Engine ID
ENG-R58848

Camera
C1 (Fixed overhead)

Correct SOP Order
B4 → B2 → B1 → B3

Station
Bolt Tightening

Bolts Monitored
4 (B1, B2, B3, B4)

Daily Target
150 cycles / day

SOP Pass Cycle

Worker follows correct bolt sequence — AI confirms pass

The AI supervisor watches the worker tighten 4 bolts in the correct SOP order (B4 → B2 → B1 → B3). Each bolt is marked with a colour-coded circle. The dashboard on the right updates live every second.

▲ ENGINE ID

▲ STATION

▲ CAMERA

▲ CYCLE TIME

▲ BOLTS DONE

ENGINE

ENG-R58848

STATION

Bolt Tightening

CAMERA

C1

CYCLE TIME

5.8s

BOLTS

1/4

SOP COMPLIANCE DASHBOARD

Live pass/fail + metrics

42% 5.8s

42%

5.8s

AI COMPUTER VISION FEED

Tracks all 4 bolt positions simultaneously

PROCESSES

1 / 1 active

Bolt Sequence

IN PROGRESS

EXPECTED

ACTUAL

42%

42%

PROGRESS

42%

target ≥ 80%

100%

CYCLE TIME

6s

target ≤ 30s

30s

CYCLES TODAY

4 / 150

Daily target: 150

- Yellow = AI actively tracking bolt

● Green = Bolt correctly tightened ✓

● Red = Wrong sequence — SOP violation ✕

EXPECTED ORDER
B4 → B2 → B1 → B3 (SOP-defined correct sequence)

ACTUAL ORDER
Filled in real time as worker completes each bolt

PROGRESS %
Cycle completion percentage — target ≥ 80%

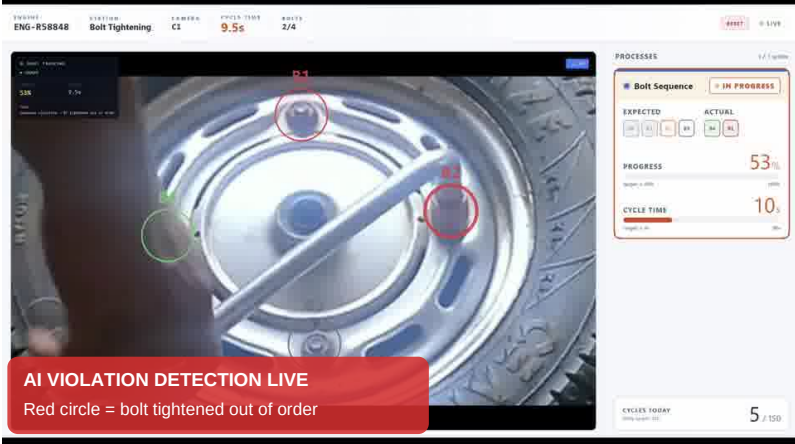
CYCLE TIME
Seconds elapsed — target ≤ 30s | **CYCLES TODAY:** daily count vs target

SOP Fail Cycle

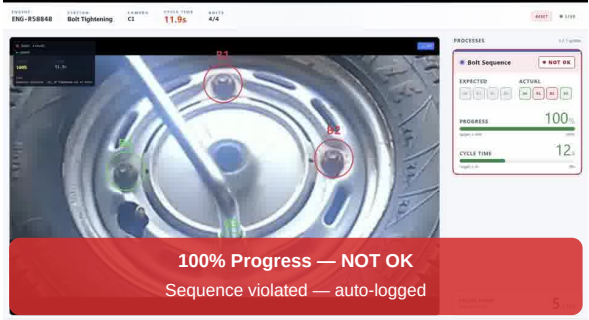
Worker tightens wrong bolt first — AI flags instantly, no human needed

Worker tightens B1 first instead of B4. AI detects the mismatch and highlights the bolt RED in under 1 second. The cycle ends as NOT OK. The sequence violation is logged automatically — no supervisor involvement required.

■ AI DETECTING VIOLATION — Wrong bolt sequence in progress



✗ CYCLE COMPLETE — NOT OK



WHAT THE AI DID IN THIS CYCLE:

- Locked onto B1 being tightened first — mismatch vs SOP (expected B4 to start)
- Highlighted B1 and B2 in RED as each out-of-sequence bolt was tightened
- Completed cycle at 100% progress but issued NOT OK verdict automatically
- Logged full cycle record: actual sequence, time taken, violation type — zero human input

PASS vs FAIL — AT A GLANCE

	SOP PASS CYCLE	SOP FAIL CYCLE
Bolt Order	B4 → B2 → B1 → B3 (correct)	B1 → B4 → B2 → ... (wrong)
Circle Colour	Yellow → Green (progress OK)	Yellow → RED (violation)
AI Verdict	IN PROGRESS → (Pass)	IN PROGRESS → NOT OK
Alert Raised	None	Instant — < 1 second
Logged	Pass cycle with timestamp	Fail cycle with violation detail

Why This Matters

The Problem

- Bolt errors cause costly field failures and recalls
- Human supervisors can't watch every station at once
- Manual QC checks are slow, inconsistent, expensive
- Sequence mistakes often caught too late in the line
- No digital audit trail of cycle-level compliance data

The BiztelAI Solution

- ✓ AI camera detects every bolt tightening event in real time
- ✓ Sequence verified in < 1 second — no human needed
- ✓ Mounts above existing station — no hardware retooling
- ✓ Instant on-screen and logged alerts on every violation
- ✓ Full digital audit trail per cycle, per shift, per station

COMPETITIVE ADVANTAGES

Plug & Play

No retooling or line modification

Edge AI

Runs on local hardware — no cloud

Multi-Process

Works for any assembly sequence

Full Audit Trail

Every cycle logged with pass/fail + time

TARGET INDUSTRIES

Automotive Assembly

Aerospace MRO

Industrial Equipment

Consumer Electronics

General Manufacturing

BiztelAI turns every factory camera into an AI quality inspector.

Real-time SOP compliance. Zero extra headcount. Instant ROI.

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